AP CALCULUS AB

Unit 1 • Lesson 1.8 notes

Lesson 1.8 • Determining Limits Using the Squeeze Theorem

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- State and apply the squeeze theorem.
- Use the limits $\lim_{x\to 0}\frac{\sin x}{x}=1$ and $\lim_{x\to 0}\frac{1-\cos x}{x}=0$.

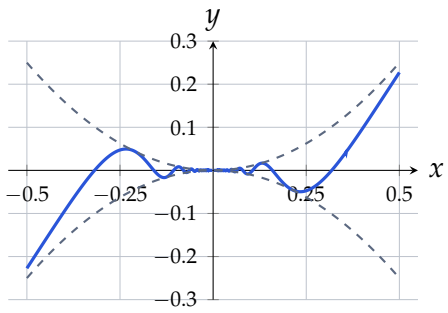
Vocabulary. squeeze (sandwich) theorem, bounded function, oscillation.

1. The squeeze theorem

Key idea

If $g(x)\le f(x)\le h(x)$ for all x near c (except possibly at c), and $\lim_{x\to c}g(x)=\lim_{x\to c}h(x)=L$, then $\lim_{x\to c}f(x)=L$.

Example 1 • A bounded factor



Solution. $-1\le \sin\frac{1}{x}\le 1$, so $-x^2\le x^2\sin\frac{1}{x}\le x^2$. Both bounds approach 0, so $\lim_{x\to 0}x^2\sin\frac{1}{x}=0$.

2. Two special trigonometric limits

Key idea

$$\lim_{x\to 0}\frac{\sin x}{x}=1 \qquad \lim_{x\to 0}\frac{1-\cos x}{x}=0 \qquad (x \text{ in radians})$$

Example 2

Solution. $\lim_{x\to 0}\frac{\sin 7x}{3x}=\frac{7}{3}\cdot\lim_{x\to 0}\frac{\sin 7x}{7x}=\frac{7}{3}.$ $\lim_{x\to 0}\frac{\tan x}{x}=\lim_{x\to 0}\frac{\sin x}{x}\cdot\frac{1}{\cos x}=1.$

Common mistakes

- Claiming $\lim_{x\to 0}x\sin\frac{1}{x}$ does not exist because $\sin\frac{1}{x}$ oscillates. The factor x forces the product to 0.
- Simplifying $\frac{\sin 7x}{3x}$ to $\frac{\sin 7}{3}$.

Quick check

1. $\lim_{x\to 0} \dfrac{\sin 2x}{\sin 5x}$ 2. If $1 - x^2 \le f(x) \le \cos x$ near 0, find $\lim_{x\to 0} f(x)$.

Answers. 1. $\frac{2}{5}$. 2. Both bounds approach 1, so 1.